13.14 Solution: The form given by Jackson is very confusing and the integral is actually a discrete sum, as $k(z)$ is piecewise constant in different media. So, I will simply write down the solution.

(a) We can write the radiation field from the polarization $\mathbf{P}(\mathbf{x}', \omega)$ in the volume element d^3x' at $\mathbf{x}'$ is

$$\begin{aligned} d\mathbf{E}_{\mathrm{rad}} &= \frac{e^{ikR}}{R}(\mathbf{k}\times\mathbf{P})\times\mathbf{k}d^3x' \\ &= \frac{e^{ikr}}{r}\left[-\frac{\omega_p^2}{4\pi\omega^2}\right]k^2\rho(z)(\hat{\mathbf{k}}\times\mathbf{E}_i)\times\hat{\mathbf{k}}e^{-i\mathbf{k}\cdot\mathbf{x}'}d^3x', \\ &= \frac{e^{ikr}}{r}\left[-\frac{\omega_p^2}{4\pi c^2}\right]\rho(z)(\hat{\mathbf{k}}\times\mathbf{E}_i)\times\hat{\mathbf{k}}e^{-i\mathbf{k}\cdot\mathbf{x}'}d^3x', \end{aligned}$$

with $\rho(0) = 1$ by convention, and $k(z) = (\omega/c)\sqrt{\epsilon(\omega, z)}$. Also, $\rho = 1$ in the foil and 0 in the air gap.

Now, as in Section 13.7, we can integrate over the foils and air gaps to obtain

$$\begin{aligned} \mathbf{F}' &= \int dx \int dy([\hat{\mathbf{k}}\times\mathbf{E}_i]_{z=0}\times\hat{\mathbf{k}})\times(\text{some phase factor in } x \text{ direction}) \\ &\quad \times \sum_{l=0}^{N-1} e^{il(p_0a+p_1b)}\int_0^a e^{ip_0z}dz, \end{aligned}$$

where

$$p_0 = \frac{\omega}{v} - k_0\cos\theta = \frac{\omega}{v} - \frac{\omega}{c}\sqrt{1-\frac{\omega_p^2}{\omega^2}}\cos\theta, \quad p_1 = \frac{\omega}{v} - k_1\cos\theta = \frac{\omega}{v} - \frac{\omega}{c}\cos\theta.$$

Comparing with single interface result in Section 13.7, we can see that the result here differs by a factor of

$$\mathcal{F} = \left|p_0\sum_{l=0}^{N-1} e^{il(p_0a+p_1b)}\int_0^a e^{ip_0z}dz\right|^2.$$

(b) By direction evaluation, we can see that

$$\mathcal{F} = \left|\frac{1-e^{iN(p_0a+p_1b)}}{1-e^{i(p_0a+p_1b)}}(1-e^{ip_0a})\right|^2 = 4\sin^2(p_0a/2)\frac{\sin^2[N(p_0a+p_1b)/2]}{\sin^2[(p_0a+p_1b)/2]}.$$

Using the result from Problem 13.13, we have

$$p_0 = \frac{\nu}{2D}\left(1+\frac{1}{\nu^2}+\eta\right), \quad p_0 = \frac{\nu}{2D}\left(1+\eta\right),$$

and

$$\mathcal{F} = 4\sin^2\Theta\frac{\sin^2[N(\Theta+\Psi)]}{\sin^2[\Theta+\Psi]},$$

with

$$\Theta = \nu\left(1+\frac{1}{\nu^2}+\eta\right)\frac{a}{4D}, \quad \Psi == \nu\left(1+\eta\right)\frac{b}{4D}.$$